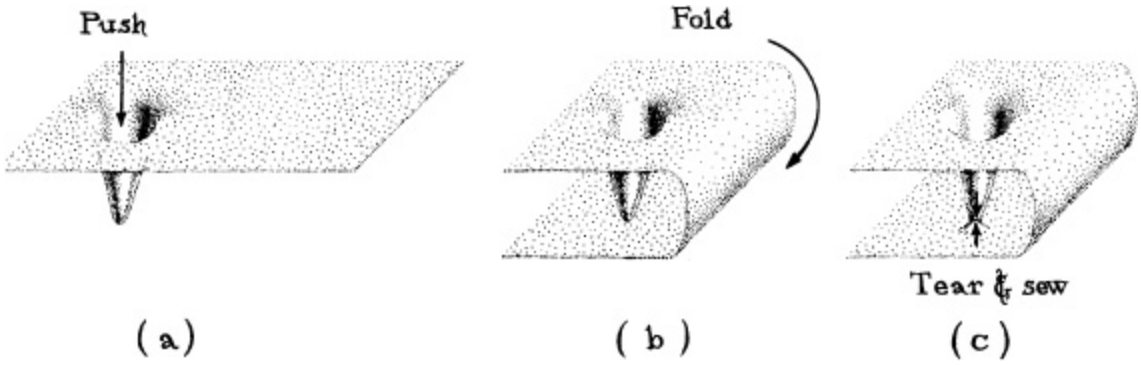
classical size. In 0.4 percent of such attempts, if the civilization were truly infinitely advanced, they might succeed. Or would they?

We do not yet understand the laws of quantum gravity well enough to know. One reason for our ignorance is that we do not understand the quantum foam itself very well. We aren't even 100 percent sure it exists. However, the challenge of this Sagan—type thought experiment—an advanced civilization pulling wormholes out of the quantum foam—might be of some conceptual help in the coming years, in efforts to firm up our understanding of quantum foam and quantum gravity.

So much for the *quantum strategy* of wormhole creation. What is the *classical strategy?*

In the classical strategy, our infinitely advanced civilization would try to warp and twist space on macroscopic scales (normal, human scales) so as to make a wormhole where previously none existed. It seems fairly obvious that, in order for such a strategy to succeed, *one must tear two holes in space and sew them together.* Figure 14.4 shows an example.



14.4 One strategy for making a wormhole. (a) A "sock" is created in the curvature of space. (b) Space outside the sock is gently folded in hyperspace. (c) A small hole is torn in the toe of the sock, a hole is torn in space just below the hole, and the edges of the holes are "sewn" together. This strategy looks classical (macroscopic) at first sight. However, the tearing produces, at least momentarily, a spacetime singularity which is governed by the laws of quantum gravity, so this strategy is really a quantum one.

Now, any such tearing of space produces, momentarily, at the point of the tear, a singularity of spacetime, that is, a sharp boundary at which spacetime ends; and since singularities are governed by the laws of quantum gravity, such a strategy for making wormholes is actually quantum mechanical, not classical. We will not know whether it is permitted until we understand the laws of quantum gravity.